Two Roads from the Discrete to the Continuum

Category-Error Avoidance and Continuum Limits Compared

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This treatise references and comments on statements from: Krüger, M. (2026). *Dimension-Graded Geometry and Typed Emergence on Aperiodic Carriers: From Cellular Degree, Holonomy, and Regge Curvature Support to a Falsifiable Spacetime-Limit Programme in an HLV Testbed.* Zenodo, DOI: 10.5281/zenodo.22105698. All direct quotations in Part 1 are drawn from this preprint or the author's accompanying summary. The refinement in Section 2.1 ("two different meanings of continuum") follows a piece of feedback from M. Krüger (IPI, 26 August 2026) on an earlier version of this document; see Doc. 190, register entries R96 and R97.

Introduction

In theoretical physics, one methodological task keeps recurring: starting from a discrete or compact structure — a lattice, a simplicial complex, a compact manifold — one wants to obtain a continuum that admits a physical interpretation (time, space, a gauge field, curvature). The passage from the discrete to the continuous is by no means automatically licensed. It requires an explicitly declared, mathematically controlled map that shows precisely *which* structure is preserved and *which* is lost. If this map is not supplied, a category error results: objects from different mathematical categories (for instance, chains of different degree in a chain complex) are tacitly identified without the identification itself being justified.

This treatise contrasts two different strategies for justifying such a passage: first, a careful, multi-stage construction within the discrete Regge-calculus and homological-lattice-vacuum tradition, which blocks the category error through explicit lemmas and *postulates* (rather than proves) the continuum limit via a refinement sequence with Mosco convergence; and second, a construction in which the continuum does not arise as the limit of a sequence but as a canonical pullback of functions along a fixed covering projection.

.1 The Discrete Construction — Chain Complexes, Regge Curvature, and the Conditional Hyperbolic Limit

The Methodological Foundation: Chain Complexes Without Physical Presupposition

The starting point is a finite oriented chain complex K with boundary maps ∂_p satisfying $\partial^2 = 0$. The dual cochain spaces $C^p = \mathrm{Hom}(C_p, \mathbb{R})$ carry coboundary operators d_p with $d^2 = 0$. Crucially, the "degree" p here is "nothing more than this chain degree — no physical meaning is built in."

Category-Error Block

$C^p \cong C^q$ for $p \neq q$ is "not well-typed unless an explicitly declared map $\Phi : C^p \to C^q$ exists that demonstrably preserves the relevant operations." The lemma "reflexively blocks 'edge = time', 'face = gauge field', '3-cell = gravity' before anyone can assert this without supplying the bridge."

Why a Single Dimension Alone Carries No Curvature

A standard result states: "in 1D the Riemann tensor is a 2-form valued in the endomorphism bundle — every 2-form on a 1-manifold vanishes trivially." Importantly, "extrinsic curvature (a line bent in $\mathbb{R}^2$ or $\mathbb{R}^3$) is something different and does not contradict the theorem" — this blocks the fallacy that a visibly curved line automatically yields a one-dimensional gravitational degree of freedom.

Edge Transport, Face Holonomy, and the Limit of Pure Gauge Transformations

A $U(1)$ phase 1-cochain θ_{ij} on edges generates, via $d\theta$, a loop phase Φ_f on faces, invariant under vertex rephasing $\theta_{ij} \mapsto \theta_{ij} + \alpha_j - \alpha_i$. If $\theta = d\alpha$ is exact, then $d\theta = d^2\alpha = 0$ — "pure vertex phase differences therefore generate no nontrivial flux on a contractible region." This marks the first point at which "a face can for the first time carry algebraically different information than an edge" — whether this ever becomes a physical gauge field remains explicitly open.

Regge Curvature Sits on Codimension Two — and the Trap of Local Tetrahedralization

In Regge calculus, curvature in a d-dimensional piecewise-flat simplicial manifold concentrates on $(d-2)$-dimensional "hinges" — in 3D on edges. The deficit angle $\varepsilon_h = 2\pi - \sum_\sigma \theta_{\sigma h}$ is "defined only if a valid global piecewise-flat complex with compatible simplex metrics exists."

The Decisive Precondition

"Local tetrahedralization of individual projected cells is not sufficient to define a global Regge curvature field if different tetrahedra overlap with positive volume or fail to meet face-to-face — in that case the star of a putative hinge is not the star of a valid manifold at all, and the deficit angle no longer has a unique geometric interpretation."

If tetrahedra overlap with positive volume (rather than satisfying the so-called HFF condition — "face-to-face": intersections of distinct closed simplices are either empty or a shared face), the geometric meaning of the deficit angle collapses, even though the individual local computations remain error-free.

This restriction concerns the classical, piecewise-flat, face-to-face Regge calculus. Generalized Regge approaches exist in the literature — for instance, constructions using simplices of constant curvature rather than flat ones [4], or allowing a variable triangulation that changes over time [5] — but these solve a different problem: they replace the flatness assumption of individual simplices, or permit a changing discretization, without altering the requirement of a globally face-to-face-consistent triangulation, and they do not address the case discussed here of overlapping simplices that fail to meet face-to-face.

Separating Algebraic Chain Structure from Geometric Realizability

A previously observed empirical failure — referred to in the preprint itself as the "007F result" — is elevated to a construction principle: a numerical investigation reported there found, according to the author, 5455 certified cases in which projected cells overlap with positive volume. An explicit distinction is drawn between the abstract provenance chain complex K_{abs} (algebraically valid solely through orientation and $\partial^2 = 0$, independent of any embedding) and the embedded realization $\iota : K_{\text{abs}} \to E_{\parallel}$, which must satisfy the strict HFF condition.

Clean Isolation of the Failure

"$\partial^2 = 0$ in K_{abs} implies $d^2 = 0$ and all finite algebraic Hodge relations — regardless of whether a concrete realization ι satisfies the strict HFF condition. What fails is only the promotion of these algebraic cells to a globally embedded manifold."

Hodge Positivity

Standard DEC statement: $\langle \omega, \Delta_p \omega \rangle = \|d_p \omega\|^2 + \|\delta_p \omega\|^2 \geq 0$, so Δ_p is positive semi-definite and self-adjoint in the declared finite inner product.

Spectral Dimension: Triviality at Both Ends

For a finite positive semi-definite operator A on a space of dimension $N = \dim \mathscr{H} < \infty$ with eigenvalues $\lambda_1, \ldots, \lambda_N$, $P(\sigma) = N^{-1} \mathrm{Tr}\, e^{-\sigma A}$ is defined, from which $d_s(\sigma) = -2\sigma P'/P$ follows.

Unavoidable Consequence

$\lim_{\sigma \downarrow 0} d_s(\sigma) = 0$ (Taylor $P \approx 1 - \sigma \bar{\lambda}$) and $\lim_{\sigma \to \infty} d_s(\sigma) = 0$ if A has a nontrivial null space. "A positive-dimensional plateau on a finite support can only be an intermediate-scale phenomenon — both endpoints are trivial."

The Continuum Commitment via Mosco Convergence — a Postulate, Not a Proof

A fixed support with bounded degree/edge weight has a bounded graph Laplacian — no intrinsic UV sequence $\lambda \to \infty$. With a shrinking scale $a_n \downarrow 0$, $A_n = a_n^{-2} L_n$, it is *postulated, not* *proved*: the Dirichlet forms $\mathscr{E}_n$ converge in the Mosco sense to $\mathscr{E}[f] = \int \nabla f^T C(x) \nabla f d^3x$ with $C(x) \approx c^2 \mathbb{1}$.

The Conditional Hyperbolic Limit

Under five unproven assumptions (Mosco convergence, uniform energy stability, convergent initial data, vanishing consistency errors, compactness), the weak limit yields $\partial_t^2 u - c^2 \Delta u = 0$ — "none of the five points is actually demonstrated in this preprint."

.2 The Alternative Construction — Continuum as Canonical Pullback Rather Than as the Limit of a Sequence

Rather than considering a sequence of discrete approximations with shrinking lattice spacing, the compact structure is treated from the outset as fundamental. The passage to the noncompact description proceeds via a **canonical function pullback** along a fixed covering projection $p : \mathbb{R} \to S^1$.

The Pullback as an Exact Map

$\Phi = p^* : C(S^1) \to C(\mathbb{R})$ is an exact, infinite-dimensional, continuous, algebra-preserving map — not a family of approximations. Every periodic function on S^1 immediately yields a periodic function on $\mathbb{R}$; the lossy direction lies at $\mathbb{R} \to S^1$, not at the compactification itself.

Why This Is Methodologically Distinct from the Category-Error Block

The category-error block is directed against identifying two chain spaces of *different degree* within a discrete chain complex. The canonical function pullback does not structurally fall into this category: it does not identify two different chain degrees, but rather two *representations of the same function algebra*. The general methodological concern remains justified for genuine chain-degree changes; it simply does not describe the case of an explicit, declared, operation-preserving pullback between function spaces.

Refinement: two different meanings of "continuum"

The pullback $\Phi = p^* : L^2(S^1_m) \to \mathrm{Per}_{R_m}(\mathbb{R}_t)$, $(\Phi f)(t) = f(t \bmod R_m)$, delivers a *continuum of the coordinate representation*: a sufficiently regular representative of f (continuous for evaluation, C^1 or a Sobolev space H^1 for differentiability) can be evaluated, differentiated, and compared against non-periodic test functions at every point $t \in \mathbb{R}$ rather than only modulo R_m. This property belongs to the representative, not to the bare L^2 element itself: $L^2(S^1_m)$ (likewise $\mathrm{Per}_{R_m}(\mathbb{R}_t)$) consists of equivalence classes of almost-everywhere-equal functions, for which a value at a fixed point t is simply not well defined without additional regularity — regardless of whether one stays on S^1_m or passes to the pullback on $\mathbb{R}_t$. It does *not* deliver a continuum of the state space: since Φf is R_m-periodic, $\int_{\mathbb{R}} |\Phi f|^2 \, dt = \infty$ for any $f \neq 0$ — Φf does not lie in ordinary $L^2(\mathbb{R}_t)$ with Lebesgue measure over the whole line. The correct target space $\mathrm{Per}_{R_m}(\mathbb{R}_t)$ (periodic L^2_{loc}, norm over one period) carries the same countable Fourier basis $\{e_n\}_{n\in\mathbb{Z}}$ and the same Parseval identity as $L^2(S^1_m)$ itself — the isometry is a change of representation within *isomorphic* Hilbert spaces, not an extension to a larger state space. Both statements — continuum of the coordinates, no continuum of the state space — are simultaneously true and do not contradict each other; they simply answer different questions. This distinction sharpens, rather than refutes, the property of the pullback as a pure change of representation described in the preceding paragraph.

The Difference from the Mosco-Convergence Strategy

The essential difference lies precisely in the "Mosco convergence" and "conditional hyperbolic limit" building blocks: there, a scaled refinement sequence with a *postulated, unproven* Mosco convergence is necessary. In the pullback construction this refinement machinery is dispensed with entirely — no sequence $a_n \downarrow 0$, no rescaled operator A_n, no Mosco commitment. The price: the compact structure itself is not derived from something more fundamental but is set as the starting point.

What the Triviality of the Spectral Dimension Means for an Operator with an Infinite Discrete Spectrum

The triviality theorem concerns explicitly a *finite* operator on a *fixed finite support*. An operator with, from the outset, infinitely many discrete eigenvalues on a compact manifold — the Laplace operator on S^1, with spectrum $\{k^2 : k \in \mathbb{Z}\}$, is a case in point — does not satisfy this precondition: $N = \dim \mathscr{H} = \infty$, and the spectrum diverges without bound rather than terminating at a finite maximum. Already the Taylor expansion $P(\sigma) \approx 1 - \sigma\bar{\lambda}$ breaks down, since a mean over infinitely many, unboundedly growing eigenvalues is not well defined. The triviality theorem therefore simply does not apply to such an operator — neither confirmed nor refuted, its precondition is absent.

.3 Summary

	Discrete chain-complex strategy	Canonical pullback strategy
Starting point	finite cell complex	compact structure set as fundamental
Category error	blocked by explicit lemma	does not arise (representation change, not degree change)
Geometric failure	precisely localized (algebra remains valid)	does not arise (no embedding question)
Continuum passage	refinement sequence, Mosco convergence *postulated*	single exact pullback, no sequence needed
Open preconditions	five unproven assumptions for hyperbolic limit	compactness of the base structure itself unexplained

Both strategies carry a price: the first pays with unproven convergence assumptions for the continuum limit; the second pays with the fact that the compactness of the fundamental structure itself stands as a starting assumption. Which price is the more acceptable one depends on the respective ambition of the construction.

.4 References

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